



VIT BHOPAL UNIVERSITY
School of Computer Science and Engineering

GradeMark Pro

Grade & Marks Tracker and Calculator

VITyarthi – Build Your Own Project
B.Tech. Computer Science and Engineering (CSE Core)
CSE1003 – Computer Problem Solving

PROJECT SUBMISSION DETAILS

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Academic Year:	2026–27

1. Introduction

GradeMark Pro is a beginner-friendly Python desktop application for recording student marks, editing or removing records, calculating grades, and viewing class statistics. The interface uses Tkinter and the project is organized into small modules suitable for a fresher to understand.

2. Problem Statement

Manual mark calculations and record keeping can cause repeated work and data-entry mistakes. GradeMark Pro combines student CRUD operations, validation, grade calculation, and basic reporting in one application.

3. Functional Requirements

- Add, edit, and delete student records.
- Validate marks from 0 to 100 and required fields.
- Prevent duplicate roll numbers.
- Display records in a table.
- Calculate percentage, grade, grade point, and PASS/FAIL.
- Show student count, average, passed and failed counts.

4. Non-functional Requirements

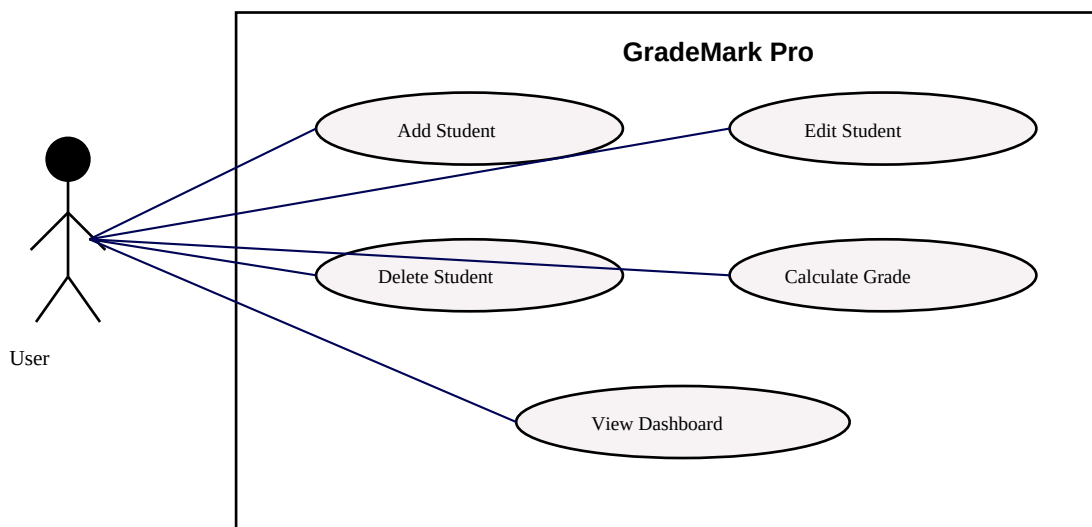
- Simple Tkinter interface.
- Input validation and error handling.
- Modular and maintainable code.
- Fast response for normal classroom-sized data.
- Uses standard Python libraries where possible.

5. System Architecture

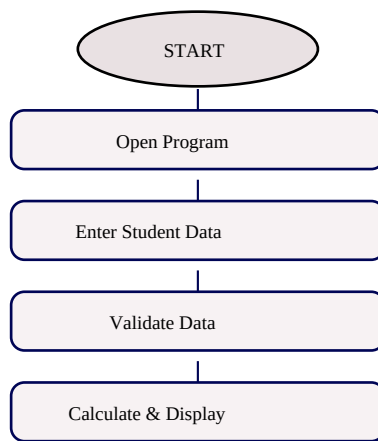
Layer	Responsibility
Presentation	Tkinter forms, buttons, table, dashboard.
Application Logic	CRUD, validation, calculations and statistics.
Data Layer	JSON file for persistent student records.
Academic Support	CSE1003 algorithm utilities and tests.

6. Design Diagrams

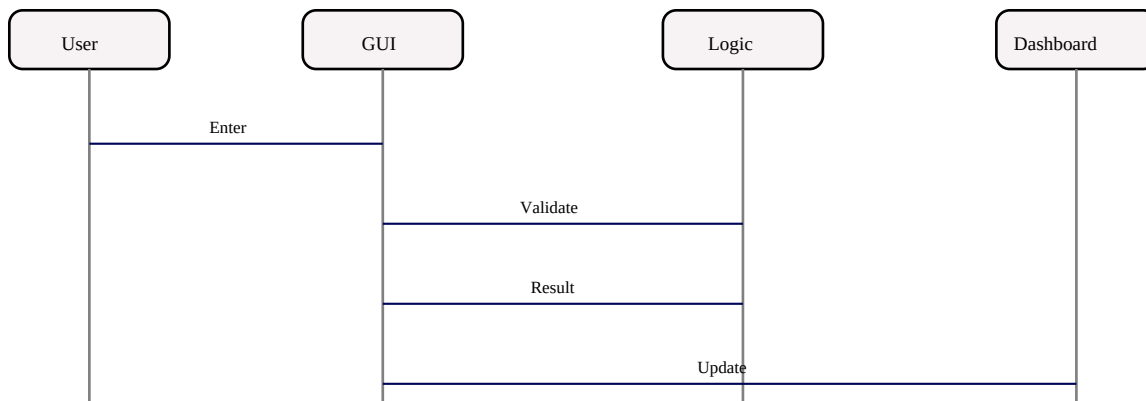
6.1 Use Case Diagram



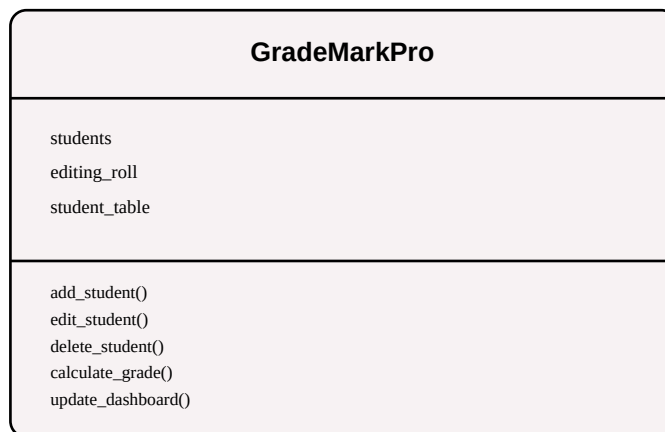
6.2 Workflow Diagram



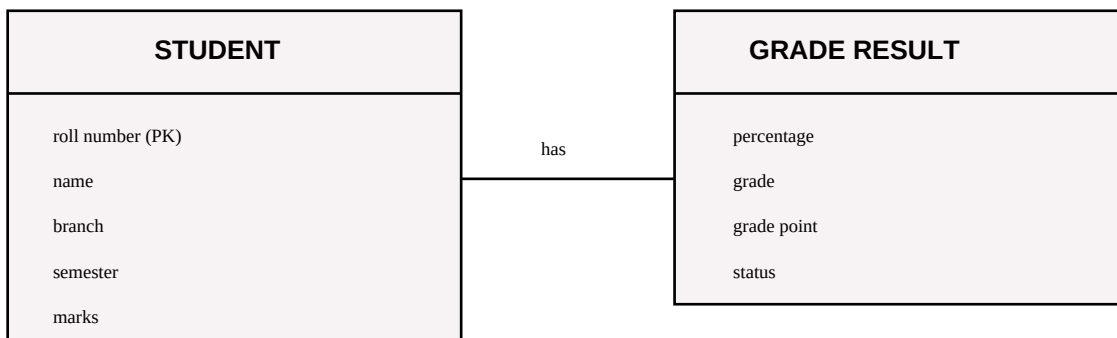
6.3 Sequence Diagram



6.4 Class / Component Diagram



6.5 ER Diagram



STUDENT and GRADE RESULT are inside their respective entity boxes.

7. Design Decisions & Rationale

Decision	Rationale
Python + Tkinter	Accessible for a fresher and suitable for a desktop GUI.
JSON storage	Simple persistent storage without database setup.
Modular structure	Each responsibility is easier to read, test and modify.
CSE1003 algorithms	Demonstrates syllabus concepts without a lengthy lab-table menu.

8. Implementation Details

- `main.py` controls the GUI and application flow.
- `student_manager.py` handles add, update, delete and lookup operations.
- `grade_utils.py` handles percentage, grades and passing logic.

- calculator.py handles individual result calculations.
- dashboard.py calculates class statistics.
- data_manager.py handles JSON persistence.
- algorithms.py contains syllabus-aligned CSE1003 algorithms.

Grade Scale

Percentage	Grade	Point
90–100	A+	10
80–89	A	9
70–79	B+	8
60–69	B	7
50–59	C	6
40–49	D	5
Below 40	F	0

Passing criterion: 40%.

9. Screenshots / Results

The project package should contain screenshots of the running Dashboard, Student Management, Add Student, Edit Marks, Delete Student, and Grade Calculator screens.

10. Testing Approach

Test	Action	Expected Result
Add	85 marks	Record appears.
Duplicate	Existing roll	Error message.
Invalid	120 marks	Rejected.
Edit	75 → 90	Record updates.
Delete	Confirm	Record removed.
Average	75 and 85	80.00%
Grade	85/100	A, 9, PASS
Fail	35/100	F, 0, FAIL

11. Challenges Faced

- Keeping GUI code manageable.
- Handling invalid input.
- Synchronizing the table and stored data.
- Supporting edit and delete safely.
- Testing academic calculations and GUI operations.

12. Learnings & Key Takeaways

- Functions, classes and modules.
- Lists and dictionaries.
- Conditions and loops.
- JSON file handling.
- GUI event handling.
- Basic algorithm implementation and testing.

13. Future Enhancements

- SQLite database.
- Subject-wise marks and CGPA.
- CSV/Excel/PDF export.
- Charts and analytics.
- Login and role-based access.
- Advanced search and filtering.

14. References

- CSE1003 course syllabus and learning material.
- Python documentation.
- Tkinter documentation.
- VITyarthi Build Your Own Project submission instructions.

15. Evaluation Rubric

The exact numerical weightage was not supplied in the provided requirements, so no numerical scores have been invented.

Area	Evidence
Problem	Problem statement and objectives.
Functionality	CRUD, calculator, dashboard and validation.
Programming	Python modules, functions, class, lists and dictionaries.
Design	Architecture, workflow, UML and ER diagrams.
Testing	Automated and manual test cases.

Area	Evidence
Documentation	README, mapping, report and checklist.